CIE AS PHYSICS (9702) • QUESTION PAPER

8.4 The Diffraction Grating

Syllabus Section: 8. Superposition **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (86 Marks)

Question 1 Easy [10 marks]

Part (a) [4 marks]

The diffraction grating equation is given by:

$$d \sin \theta = n\lambda$$

State the definition of the following variables and an appropriate unit for each.

(i) d

[1]

(ii) θ

[1]

(iii) n

[1]

(iv) λ

[1]

Part (b) [2 marks]

A diffraction grating has 720×10^3 lines per metres.

Calculate the distance between adjacent slits of the grating.

Part (c) [3 marks]

A narrow beam of light of wavelength 635 nm is incident normally on the diffraction grating, as shown in Fig. 1.1

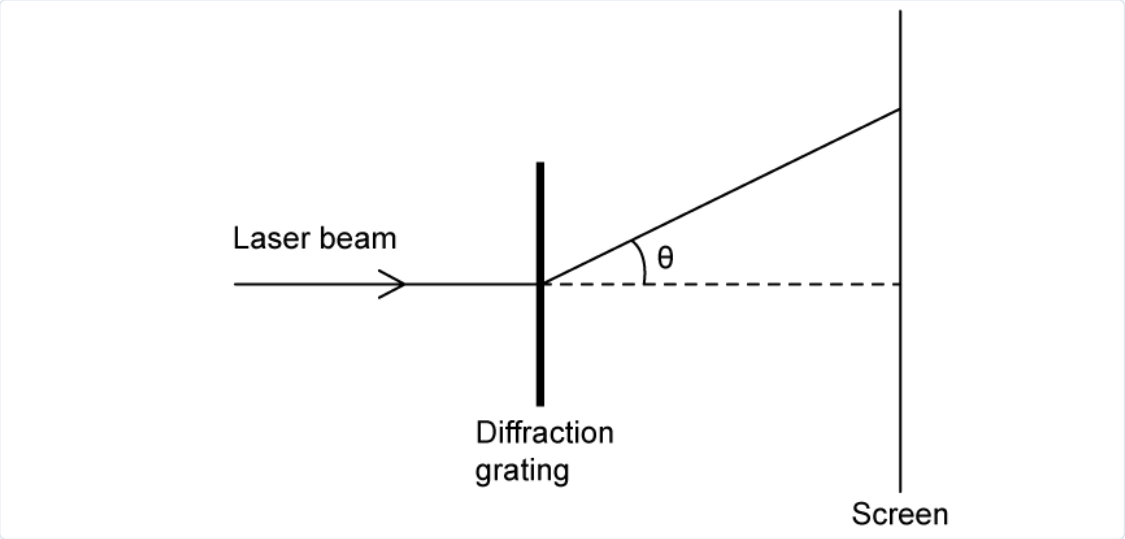


Fig. 1.1

The first-order diffraction maximum of the light is at an angle of θ to the direction of the incident light beam.

Calculate the value of θ.

Part (d) [1 mark]

The highest order observed on the screen is the **second-order** diffraction maximum.

State how many minima are seen on the screen.

Question 2 Easy [8 marks]

Part (a) [2 marks]

In a laboratory experiment, monochromatic light from a laser of wavelength λ is incident normally on a diffraction grating.

The diffracted laser light is projected on a white screen which is parallel to the plane of the grating. Fig 1.1 shows the position of the diffraction maxima.

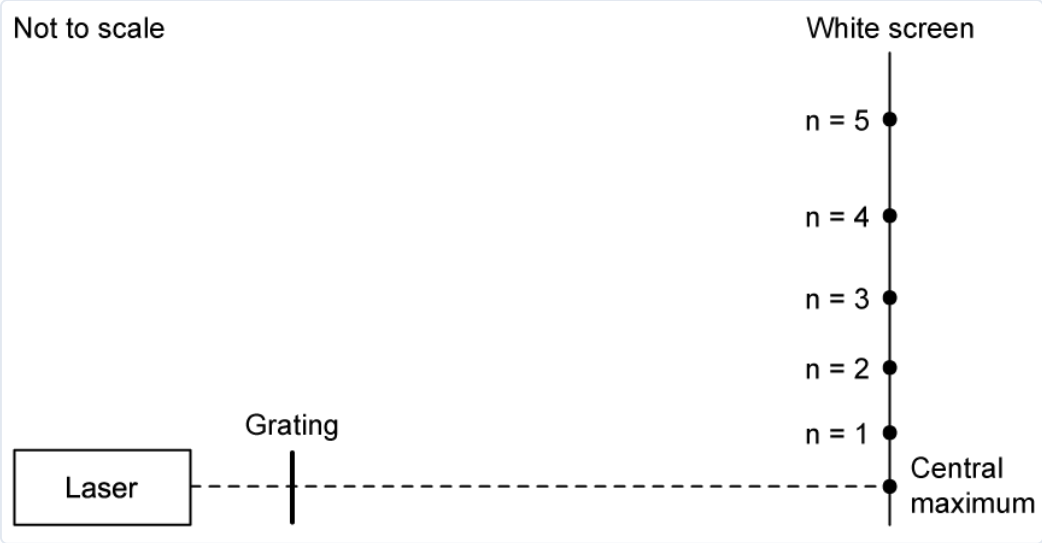


Fig. 1.1

On Fig. 1.1, sketch the diffracted laser light and mark the second and fifth-order diffraction maxima angles, θ_2 and θ_5 respectively.

Part (b) [2 marks]

The spacing between each slit in the grating is 2.45 μm.

Calculate the number of lines per metre in this grating.

Part (c) [3 marks]

The third–order spectral line is formed at an angle of 36° from the normal to the grating.

Calculate the value of λ.

Part (d) [1 mark]

State the order that would produce the brightest image on the white screen.

Question 3 Medium [14 marks]

Part (a) [3 marks]

A teacher explains a diffraction investigation to a group of physics students.

'In this investigation a laser will be used to shine monochromatic light of wavelength 581.9 nm onto a diffraction grating. The light passing through the grating will therefore be coherent.'

Define the following terms used by the teacher

(i) Monochromatic

[1]

(ii) Coherent

[2]

Part (b) [6 marks]

During the investigation, a second-order maximum is produced at an angle of 35° measured from the normal to the grating.

(i) Calculate the number of lines per metre on the grating.

[3]

(ii) Calculate the highest order which is observable.

[3]

Part (c) [3 marks]

Calculate the angular separation between the highest order and second order maxima.

Part (d) [2 marks]

The investigation is repeated using the same grating but a different monochromatic source.

In this experiment, the second-order maximum is observed at an angle of 25.7°.

Calculate the wavelength of this second source.

Question 4 Medium [11 marks]

Part (a) [3 marks]

A laser produces a monochromatic light of wavelength 581.9 nm which falls normally on a diffraction grating. A second order maximum is produced at an angle of 35° measured from the normal to the grating.

Calculate the number of lines per metre on the grating.

Part (b) [3 marks]

Calculate the highest order which is observable.

Part (c) [3 marks]

Calculate the angular separation between the highest order and second order maxima.

Part (d) [2 marks]

When the same grating is used with a different monochromatic source, the second-order maximum is observed at an angle of 25.7°.

Calculate the wavelength of this second source.

Question 5 Medium [11 marks]

Part (a) [4 marks]

A helium-neon laser produces monochromatic light of wavelength 672 nm which falls perpendicular on a diffraction grating which has 350 lines per mm.

Calculate the angle between the second-order maxima.

Part (b) [3 marks]

Show that the highest order image shown in this arrangement is n = 4.

Part (c) [2 marks]

A different laser that produces monochromatic light of wavelength λ falls perpendicular to a diffraction grating from which adjacent lines are 5λ apart.

Calculate the angle for the third order maximum.

Part (d) **[2 marks]**

The diffraction grating is placed 2.7 m from a screen.

Hence or otherwise, calculate the distance between the third diffraction order and the normal to the grating as measured on the screen.

Question 6 Hard [15 marks]

Part (a) [5 marks]

Students in a laboratory have created the following set-up shown in Fig 1.1. Parallel rays of monochromatic light from two adjacent slits A and B of wavelength λ are incident normally on a diffraction grating with a slit separation d.

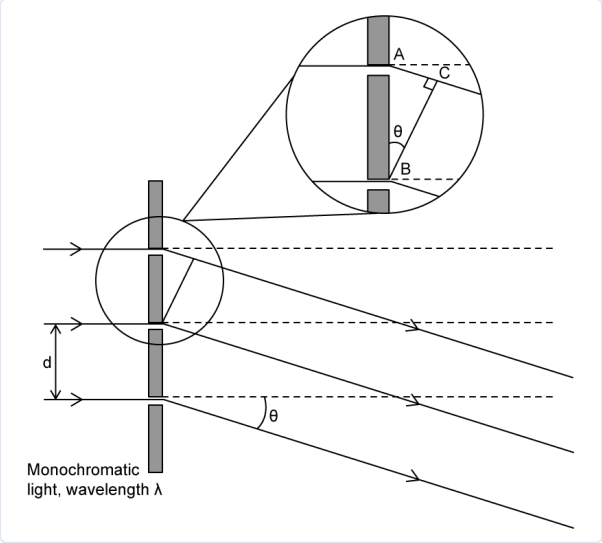


Fig. 1.1

Use the diagram to derive the equation $n\lambda = d\sin\theta$ where θ is the angle of diffraction of a maxima order n visible on a screen.

Part (b) [5 marks]

The monochromatic light in the set-up in part (a) has a wavelength of 499 nm. The graph in Fig. 1.2 shows the variation of $\sin\theta$ with the order n of the maximum. The central order corresponds to $n = 0$.

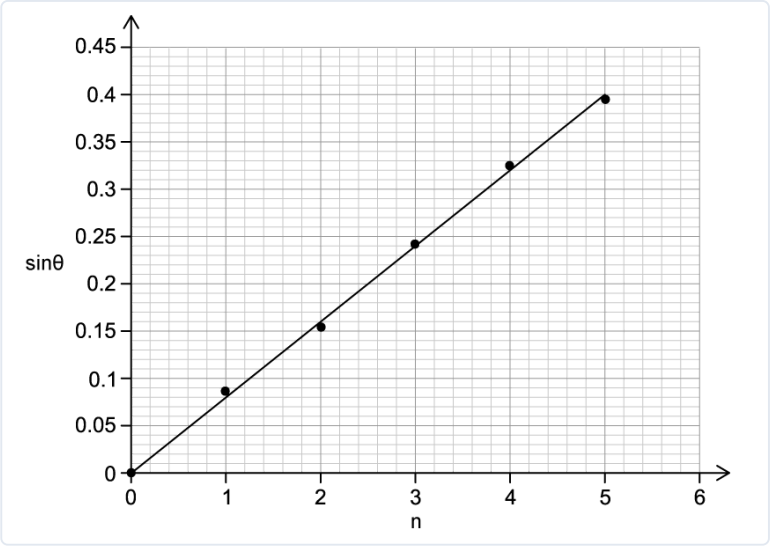


Fig. 1.2

Determine a mean value for the number of slits per mm of the grating.

Part (c) [3 marks]

The grating is 30 mm wide.

Determine the number of slits required to obtain a maximum of five bright fringes on the screen.

Part (d) [2 marks]

The students claim that they observed the following diffraction pattern in Fig. 1.3 on the screen for the grating from part (c).

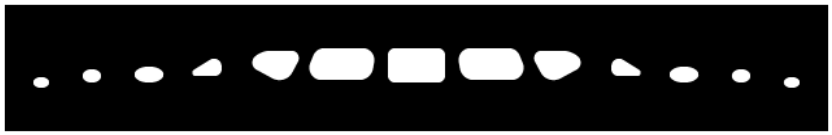


Fig. 1.3

State two reasons why the interference pattern obtained cannot be correct.

Question 7 Hard [8 marks]

Part (a) [3 marks]

A narrow beam of coherent light of wavelength 520 nm is incident upon a diffraction grating that has 3.10×10^2 lines per mm. The diffraction pattern is seen on a screen.

Calculate the maximum number of minima that could be seen on the screen.

Part (b) [2 marks]

A different light source is incident upon the same diffraction grating. A student suspects there are two wavelengths of light in this incident beam, one of wavelength 490 nm and another of wavelength 490.2 nm.

Determine the highest order of the visible diffracted light.

Part (c) [2 marks]

The minimum angular separation for which the two wavelengths may be differentiated is 0.05°.

Deduce whether the student observed the two wavelengths as distinct images at this order. Support your answer with calculations ignoring the blurring effect often observed at higher order maxima.

Part (d) [1 mark]

Fig. 1.1 shows the shortest wavelength of this light incident on the screen after the other has been removed from part c).

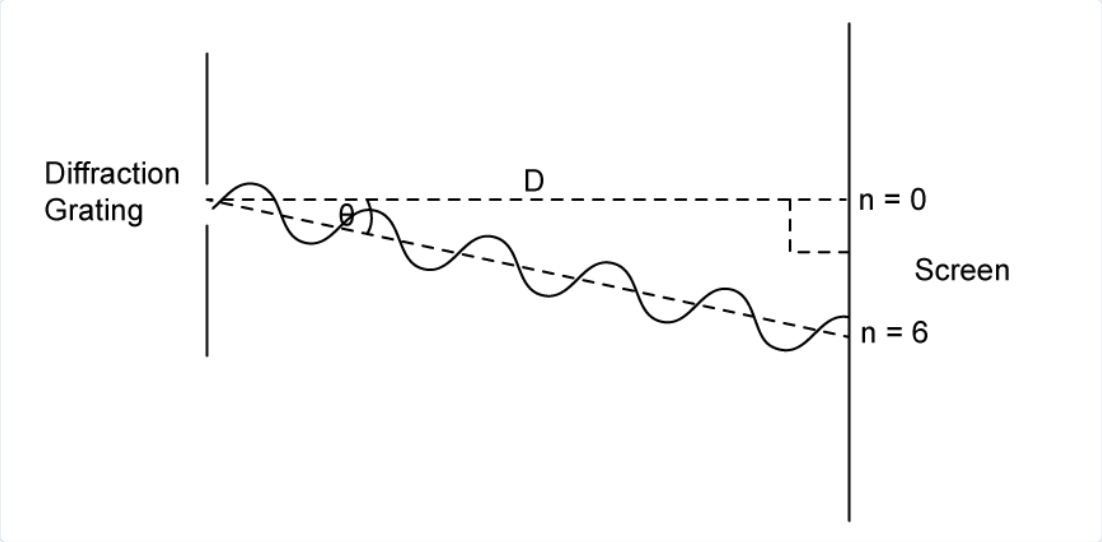


Fig. 1.1

Determine the distance between the diffraction grating and the screen, *D*.

Question 8 Hard [9 marks]

Part (a) [2 marks]

Monochromatic light is incident normally on a diffraction grating.

Describe the diffraction of the light waves at the grating.

Part (b) [4 marks]

A diffraction pattern forms on a screen.

The light has a wavelength of 630 nm. The diffraction grating has 4.50×10^2 lines per millimetre.

Determine the total number of gaps between orders, formed on the screen.

Part (c) [3 marks]

The beam of light in **(b)** is replaced by a beam of blue light incident on the same diffraction grating.

(i) State whether the frequency of the blue light is greater than, less than, or the same as the frequency of the original light.

[1]

(ii) State and explain the effect of this change on the number of gaps formed on the screen. A calculation is not required.

[2]